TIC 200247352

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        3.337797 d
  Epoch (t0)    1659.3227 BTJD
  Duration      0.0840 d
  Depth         2.671 %  (26,711 ppm)
  SNR           413.1

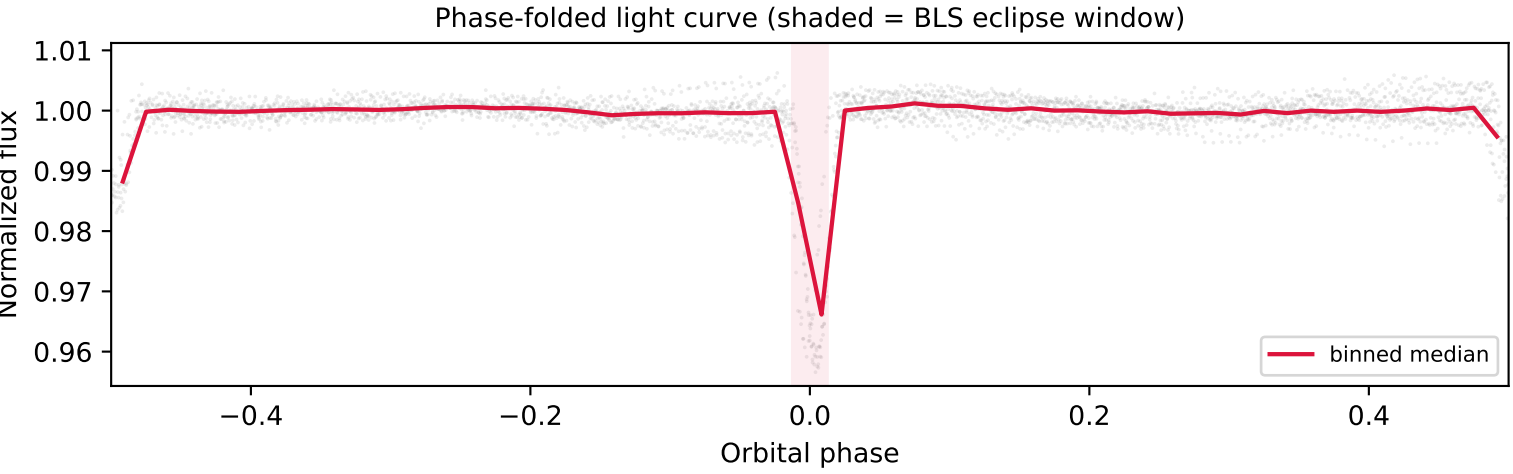
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      13  (gate >= 3)
  Robust depth ratio     1.18  (gate >= 0.50)
  Robust depth sigma     154.7  (gate >= 7.0)
  Odd/even z-score       0.84  (consistent)
  Secondary eclipse      detected, phase -0.49, 8.1 sigma
```

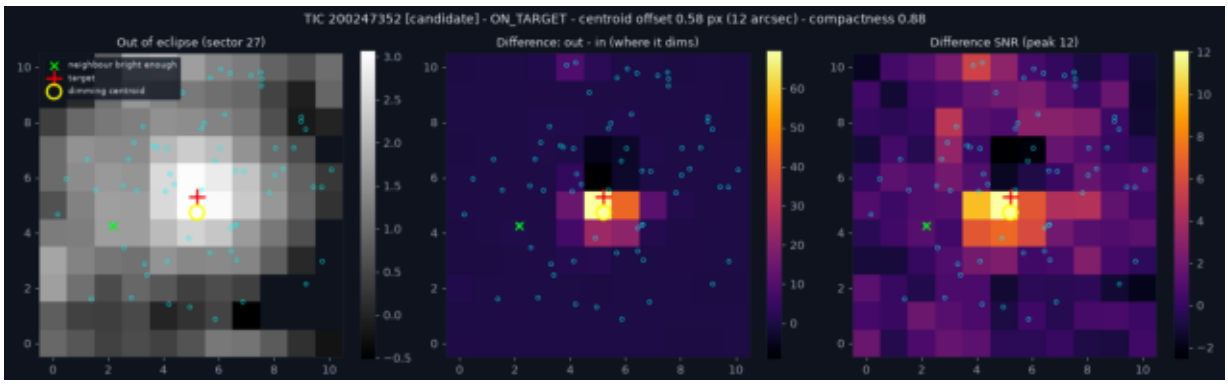
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CATALOG CROSS-MATCH (novelty)

  VSX                                  no match
  ASAS-SN                              no match
  Gaia eclipsing-binary table          no match
  Gaia variable classification         no match
  TESS EB catalog (Prsa+2022)          no match
  ExoFOP TOI list                      no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 27    light-curve dip in this sector: 139.3 sigma    t0 refined by 0.0025 phase
  Difference-image peak SNR   12.08   (gate >= 5.0)      Compactness   0.88   (gate >= 0.50)
  Dimming centroid offset     0.58 px (12 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 200247346
    3.09 px away, dTmag +3.40 - farther from the dimming than the target
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No caveats - clean detection on every check.